Math 534H – Spring 2026 – Lecture 11

In Strauss one your exercises asks find the solution to IVP

$$\begin{cases} u_{xx} - 3u_{xt} - 4u_{tt} = 0 \\ u(x,0) = x^2 \\ u_t(x,0) = e^x \end{cases}$$

Short sketch of what you need to do:

1. Complete squares $\longrightarrow \underbrace{\left(\partial_x - \frac{3}{2}\partial_t\right)}_{\partial_\xi}{}^2 - \underbrace{\left(\frac{5}{2}\partial_t\right)}_{\partial_\eta}{}^2 = 0$

2. Change of Coordinates

$$\partial_\xi = \partial_x - \frac{3}{2}\partial_t$$

$$\partial_\eta = \frac{5}{2}\partial_t \qquad (x,t) \to (\xi,\eta)$$

$$x = \alpha\xi + \beta\eta$$

$$t = \gamma\xi + \delta\eta \qquad \ldots\ldots$$

$$(\partial_x - 4\partial_t)(\partial_x + \partial_t)u = 0$$

Solution should be

$$u(x,t) = F(x-t) + G(4x+t)$$

3 After change of coordinates and <u>undoing it</u>

$$\left(\begin{array}{c} \partial_{\xi\xi} u - \partial_{\xi\xi} u = 0 \\ \underbrace{\underline{\underline{f}}(\xi+\eta) + \underline{\underline{g}}(\xi-\eta)} \end{array} \right)$$

$$\underline{(\xi, \eta) \longrightarrow (x, t)}$$

you get $\quad u(x,t) = F(x-t) + G(4x+t)$

4 Now need to find F, G for the given initial conditions

Let's rewrite $u(x,t) = F(x-t) + \tilde{G}(x + t/4)$

$F, \tilde{G} : \mathbb{R} \to \mathbb{R}$ diff. fcs

$$x^2 = u(x,0) = F(x) + \tilde{G}(x)$$

Then $u_t(x,t) = -F'(x-t) + \frac{1}{4}\tilde{G}'(x + t/4)$

$$e^x = u_t(x,0) = -F'(x) + \frac{1}{4}\tilde{G}'(x)$$

2 eqs in x

We get $\begin{cases} x^2 = F(x) + \tilde{G}(x) \\ e^x = -F'(x) + \frac{1}{4}\tilde{G}'(x) \end{cases}$. Find $\underline{F, \tilde{G}}$

5 Differentiate w.r.t x this equation to get

$$\begin{cases} 2x = F'(x) + \tilde{G}'(x) \\ e^x = -F'(x) + \frac{1}{4}\tilde{G}'(x) \end{cases}.$$

6 You solve for F, $\tilde{G}$ and use initial position to determine the constants

can always check this $u(x,t)$ satisfies IVP

7 In the end you get

$$u(x,t) = x^2 + \frac{t^2}{4} + \frac{4}{5}\left(e^{x+t/4} - e^{x-t}\right)$$

Causality and Energy

FINITE PROPAGATION SPEED

Recall D'Alambert's formula

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 \\ u(x,0) = \phi(x) \in C^2 \\ u_t(x,0) = \psi(x) \text{ at least } C^1 \end{cases}$$

$$u(x,t) = \frac{1}{2}\left[\phi(x+ct) + \phi(x-ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(y)\,dy$$

Remark: Note that if $\phi \in C^2$ and $\psi \in C^1$ then D'Alambert formula for $u(x,t)$ gives that the solution is also $C^2(\underset{x}{\mathbb{R}} \times \underset{t}{[0,\infty)})$ and $\boxed{\text{NO BETTER!}}$

Wave eq. has <u>no</u> regularization effect.

<u>Causality</u> (Wave)

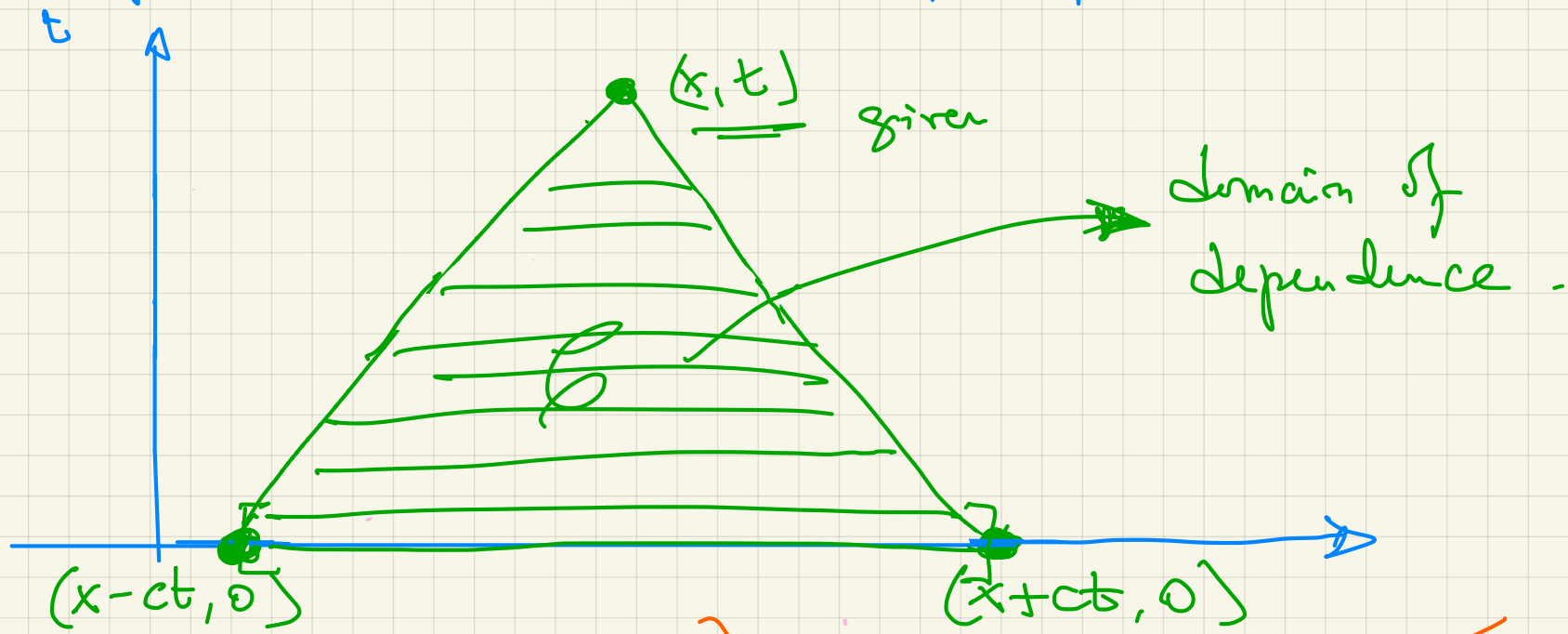
- Finite propagation
- Domain of dependence
- Domain of influence

① Effect of the initial position yield a pair of waves travelling in either direction at speed EXACTLY <u>c</u> (and at ½ of the original amplitude)

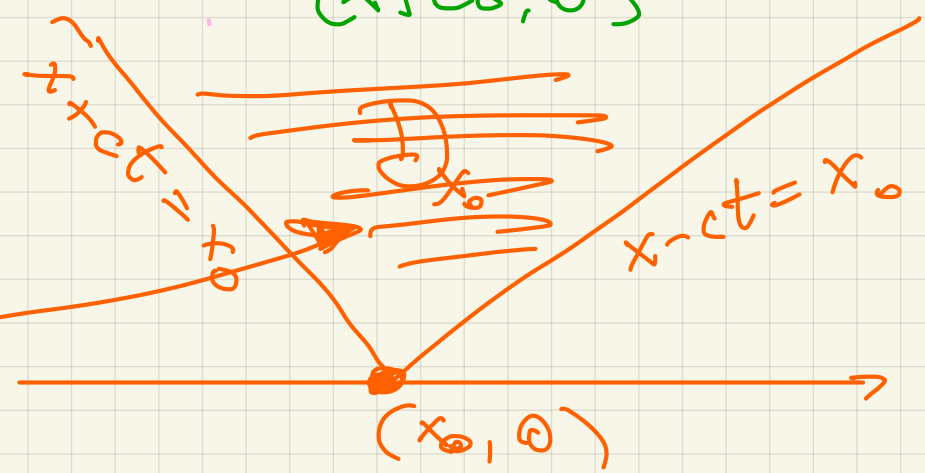
② Effect of the initial velocity yields a wave that propagates out at speed $\leq c$. So part of the <u>wave</u> may lag behind if $\psi \not\equiv 0$
↳ initial velocity.

But ① + ② $\Rightarrow$ <u>no</u> part of the wave moves <u>Faster than c</u>

Causality pple says that a wave at a point (x,t) in $\mathbb{R}\times[0,\infty)$ is only affected by the initial conditions within a past domain (cone) of dependence $\mathcal{C}$:



Domain of influence



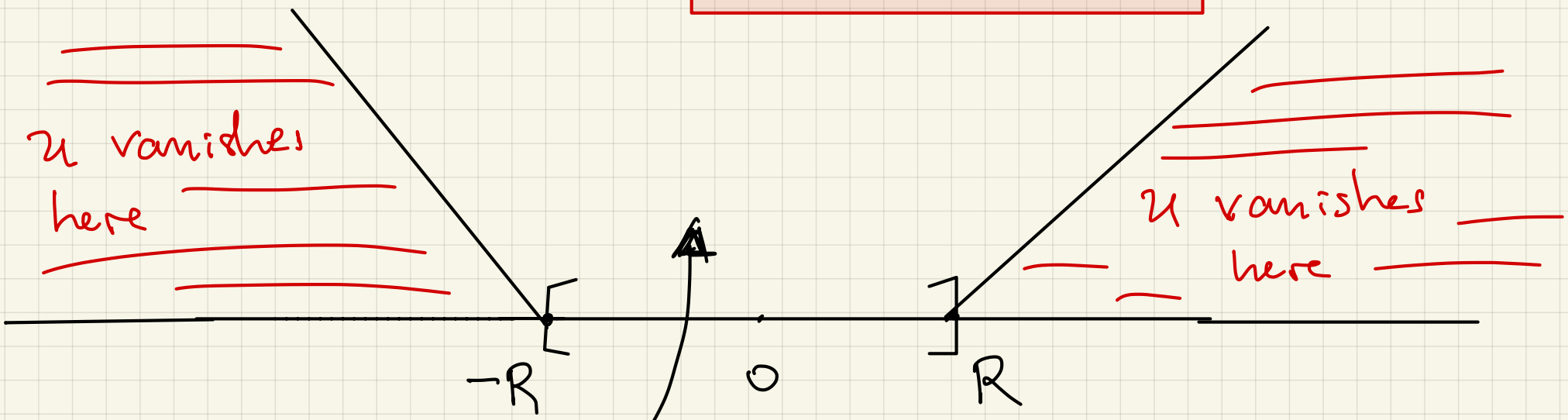
- The Domain of Dependence means that the value of u at a given point (x,t); that is $u(x,t)$ depends on the values of ϕ at the points in the x-axis $(x-ct, 0)$ and $(x+ct, 0)$ and on the values of ψ over the whole interval $[x-ct, x+ct]$

- The domain of influence of a point $x_0 \in \mathbb{R}$: initial conditions at $\underline{x_0}$ can only affect the solution $u(\underline{x,t})$ – assume $t>0$ – <u>for (x,t) ONLY in</u>

$$\underline{D_{x_0}} := \{ (x,t) \,/\, |x-x_0| \leq ct \}$$

(this cone is within the lines $x+ct=x_0$ and $x-ct=x_0$ as in the picture in previous page)

As a consequence, if the <u>initial conditions ϕ, ψ vanish</u> for $|x| > R$, then $u(x,t) = 0$ for all (x,t) such that $\boxed{|x| > R + ct}$:



Note that this region is the union of all the cones of influence with $x_0 \in [-R, R]$. This sector is $|x| \leq R + ct$

<u>Theorem</u> (Finite propagation speed): If ϕ and ψ are zero on an interval in the x-axis $[x_0 - ct_0, x_0 + ct_0]$ then $u(x_0, t_0) = 0$

<u>Remark</u> : Going back to the Domain of Dependence we get that initial conditions (information or a disturbance) outside the interval $[x-ct, x+ct]$ has no influence on the value of u at (x,t).

- All in all the information we described above give us the following theorem :

<u>Theorem</u> (Finite propagation speed) : If ϕ and ψ are zero on an interval in the x-axis $[x_0-ct_0, x_0+ct_0]$ then $u(x_0,t_0)=0$

- Next time we'll discuss <u>Conservation of Energy</u> for the wave equation.